

website\_analytics.py X

website\_analytics.py

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 df = pd.read_csv("website_data.csv")
5
6 print("Website Analytics Data")
7 print(df)
8
9 print("\nTotal Visitors:", df["Visitors"].sum())
10
11 print("Average Visitors:", df["Visitors"].mean())
12 print("Highest Visitors:", df["Visitors"].max())
13
14 print("Average Bounce Rate:", df["BounceRate"].mean())
15
16 highest_day = df.loc[df["Visitors"].idxmax()]
17 print("\nHighest Traffic Day:")
18 print(highest_day)
19
20
21 plt.plot(df["Date"], df["Visitors"], marker="o")
22 plt.title("Website Visitors Trend")
23 plt.xlabel("Date")
24 plt.ylabel("Visitors")
25 plt.xticks(rotation=45)
26 plt.show()
27
28 plt.bar(df["Date"], df["PageViews"])
29 plt.title("Page Views by Date")
30 plt.xlabel("Date")
31 plt.ylabel("Page Views")
32 plt.xticks(rotation=45)
33 plt.show()
34
35
36 df.to_csv("processed_website_data.csv", index=False)
37
```

LEARN PYTHON 3 - CHAPTER 4 - PLOTTING WITH MATPLOTLIB

```
PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection> python website_analytics.py
```

Website Analytics Data

|   | Date       | Visitors | PageViews | BounceRate |
|---|------------|----------|-----------|------------|
| 0 | 2025-01-01 | 1200     | 3500      | 45         |
| 1 | 2025-01-02 | 1500     | 4200      | 40         |
| 2 | 2025-01-03 | 1800     | 5000      | 38         |
| 3 | 2025-01-04 | 1700     | 4800      | 42         |
| 4 | 2025-01-05 | 2000     | 5500      | 35         |

Total Visitors: 8200

Average Visitors: 1640.0

Highest Visitors: 2000

Average Bounce Rate: 40.0

Highest Traffic Day:

Date 2025-01-05

Visitors 2000

PageViews 5500

BounceRate 35

Name: 4, dtype: object

Processed file saved successfully

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\veera\OneDrive\Desktop\Task1_DataCollection> python website_analytics.py
```

```
Total Visitors: 8200  
Average Visitors: 1640.0  
Highest Visitors: 2000  
Average Bounce Rate: 40.0
```

Highest Traffic Day:

```
Date      2025-01-05  
Visitors  2000  
PageViews 5500  
BounceRate 35  
Name: 4, dtype: object  
[]
```



